Kohn-Sham model of dirac16complex in the author's primordial field with the deflating extra times

The block reduction, the instantaneous ground and first excited states along the prescribed history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$, Mermin thermodynamics, adiabaticity, the energy-momentum tensor, the rescaling partners and the source conditions, with the verification records and an exact statement of what is computed and what is not

dirac16complex project (with Claude Opus 5.5)

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Abstract

This document records the Kohn-Sham (density-functional) model of the fermion field dirac16complex in the author's primordial gravitational field, deliverable 4 of `Revision/SPEC.md` section 10, as computed by the code in `Revision/kohn_sham/`, in the corrected metric of 2026-10-16 (version 2): $g = \mathrm{diag}\big(e^{-2a_4}b_4\cot^2 z,\ e^{2a_4}\ (\times 3),\ -1,\ -e^{-2a_4}b_4(\sin^2 z)^{1/6}\ (\times 3)\big)$, $z = 6Hx_0$ real, with two functions $a_4(x_4)$ and $b_4(x_4) > 0$ (b_4 is never 1). Exact (two independent verifiers, WolframScript and sympy): in the good sector (no extra-time momentum) the ansatz $\Psi = e^{ik\cdot x}e^{-(3/2)Hy}e^{a_4/2}b_4^{-1}\chi(y, x_4)$ removes the spin-connection terms $\frac{3}{2}(H/f)\gamma^0$ and $(b_4'/b_4 - \frac{1}{2}a_4')\gamma^4$ and reduces the 16-component equation, for every history $a_4(x_4)$, $b_4(x_4)$, to eight 2×2 blocks $h_j = j[-(i/f)\sigma_1\, d/dy + M_{\mathrm{eff}}\sigma_2 + \kappa k\sigma_3] + v_v$ in the hidden coordinate $y = \ln|\sin z|/(6H)$, with the hidden scale $f = e^{-a_4}b_4^{1/2}$ (the hidden direction shrinks together with the extra times) and the constant momentum weight $\kappa = e^{-a_4}$; the deflation enters the instantaneous problem through f, through the redshift κk of the 3-momenta and through the proper cell volume, which grows like $e^{-a_4}b_4^2$, on the history like $e^{(5/3)a_4}$; the exact local exchange of the uniform good-sector gas gives $M_{\mathrm{eff}} = m + \frac{15}{16}\lambda S$ and $v_v = -\lambda n/16$. Computed (a Rust solver, an independent Python reference with a different discretisation, and a cross-check of the full canonical matrix, 31 of 31 checks PASS over 217571 comparisons): the instantaneous ground and first excited states of 75 parameter sets ($N = 8, 592, 2352$; five couplings $\hat{\lambda}$, held fixed while the cell grows, so that the effective coupling falls like $e^{-(5/3)a_{4,0}}$; five slices $a_{4,0}$ of the prescribed history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$, $A = 1$), 135 Mermin states at three temperatures, the adiabaticity measure ($Q_{\max} \le 0.1982205$ over the matrix), the Fermi-level crossings (none in the canonical matrix; a demonstration at $N = 2360$), the energy-momentum tensor profiles, the exact rescaling partners and the exact-exchange variant. The free states of the matrix are exact fold-band gases: their occupied levels are $\varepsilon = \kappa|k|$ with the orbital $(e^{fMy}, 0)$, so their energy redshifts exactly like e^{-a_4}, with $\int p_3 = E_{KS}/3$ and $\int p_0 = \int p_t = 0$. ASSUMED or chosen: both fold conditions filled at the fold $z = \pi/2$ where the two halves of a cell meet, the regular tip at $y = -3$, the filling convention. The dependence on the tip cutoff is measured ($L = 3$ to 6): the free fold-band gases do not depend on it (except $N = 2352$ at $a_{4,0} = 0$), the interacting zero modes and the $k = 0$ bulk levels do (section 5). Not admissible as sources: no recorded Kohn-Sham state satisfies the source conditions of the a_4 and b_4 equations, so the history is a prescribed background without back-reaction. Not done: the time-dependent (non-adiabatic) Kohn-Sham problem, correlation beyond exchange, back-reaction.

1 The task and the answer

The author asked (2026-10-01, quoted from `Revision/README.md`): "plan and employ a DFT-motivated approximation similar to the one that you already created [...], with a Kohn-Sham fermion-gas thermodynamic effective potential, that employs the DFT ground and first-excited-state computation". `Revision/SPEC.md` section 7 makes this precise: a Kohn-Sham fermion-gas model of dirac16complex in the good sector with the contact interaction $U = \frac{\lambda}{2}S^2$, the Hartree mean field plus the exact local exchange of the uniform gas, Mermin's finite-temperature functional, the reduction to 2×2 blocks in the hidden coordinate, the boundary conditions, the ground and first excited states, the thermodynamics, and, because a_4 and b_4 depend on x_4, the INSTANTANEOUS (adiabatic) Kohn-Sham states along the history with the extra times deflating, with an adiabaticity measure.

The answer of this record:

1. Exact (section 3, section 4): the block reduction, the exchange functional, the boundary conditions, the rescaling identity, the energy-momentum tensor and its conservation laws, the

adiabaticity measure, all for the corrected metric. Every formula is in `Revision/kohn_sham/ks-theory.json` and verified twice, in `Revision/kohn_sham/reports/ks-theory-wolfram.json` (51 of 51 PASS) and independently in `Revision/kohn_sham/reports/ks-theory-python.json` (61 of 61 PASS).

2. Computed (sections 5 to 10): the instantaneous Kohn-Sham states of the canonical matrix by the Rust solver, independently by the Python reference, and compared state by state on the full matrix.

3. Not admissible (section 11): the recorded states are not sources of the field equations for a_4 and b_4; the history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$ is PRESCRIBED.

4. Open (section 15): the time-dependent Kohn-Sham problem, correlation, back-reaction.

Every number below is read from the named Revision report or results file (the publication test re-reads them); units: $H = m = 1$, energies, momenta and temperatures in units of m, as in `Revision/kohn_sham/results/parameters.json`. Interpretations are labelled.

2 Setting: the author's field, the good sector and the hidden coordinate

The coordinates are the author's, $x_0, x_1, \ldots, x_7$, in the author's order: x_0 is the hidden direction, space-like, $z = 6Hx_0$ real; x_1, x_2, x_3 are ordinary 3-space, inflating with the scale factor e^{a_4}; x_4 is the time; x_5, x_6, x_7 are the three extra times, which DEFLATE exponentially, with the scale factor $e^{-a_4}b_4^{1/2}(\sin^2 z)^{1/12}$ and a_4 increasing (they are never treated as static); the hidden direction has the scale factor $e^{-a_4}b_4^{1/2}|\cot z|$ and shrinks together with them. The metric of `Revision/SPEC.md` section 1 (version 2 of 2026-10-16; `Revision/metric_warped/`) is

$$\begin{aligned} ds^2 = {} & e^{-2a_4(x_4)}b_4(x_4)\cot^2 z\, dx_0^2 + e^{2a_4(x_4)}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \\ & - e^{-2a_4(x_4)}\, b_4(x_4)(\sin^2 z)^{1/6}\,(dx_5^2 + dx_6^2 + dx_7^2); \end{aligned}$$

$b_4(x_4) > 0$ is a function of the time and never 1. The history of this record is

$$a_4 = AHx_4, \qquad b_4 = b_4(0)\, e^{\beta AHx_4}, \qquad A = 1, \quad \beta = \tfrac{4}{3},$$

PRESCRIBED, not solved for: the record keeps the history of `Revision/SPEC.md` section 1 for continuity, and in the corrected metric it needs an anisotropic, x_0-independent source with $p_0 = p_t$ that grows with x_4 (`ks-theory.json`, key adiabaticity.historyBetaSource; section 11). On it the hidden scale is $f = e^{-a_4}b_4^{1/2} = e^{-a_4/3}$ ($b_4(0) = 1$).

The Kohn-Sham record uses the hidden coordinate $y = \ln|\sin z|/(6H) \in (-\infty, 0]$, with $dy = \cot z\, dx_0$, on every quarter of the period of z; the fold $\cos z = 0$ ($z = \pi/2$) is $y = 0$ and the tip $\sin z \to 0$ is $y \to -\infty$; the two halves $z < \pi/2$ and $z > \pi/2$ of a cell $(0, \pi)$ form the doubled system. In it the metric is exactly

$$ds^2 = e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 + f^2\big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2)\big], \qquad f = e^{-a_4}b_4^{1/2},$$

with the warp $W = e^{Hy}$ on the extra times only (3-space has no warp) and $\sqrt{|g|} = e^{-a_4} b_4^2 e^{3Hy}$ ($= e^{-a_4} b_4^2 |\cot z| (\sin^2 z)^{1/4}$ in the author's chart). The proper 7-volume of a coordinate cell grows like $e^{-a_4} b_4^2$, on the history like $e^{(5/3)a_4}$. The canonical spin connection gives $\gamma^\mu \Omega_\mu = \frac{3}{2}(H/f)\gamma^0 + (b_4'/b_4 - \frac{1}{2}a_4')\gamma^4$ for every $a_4(x_4)$, $b_4(x_4)$: the hidden direction contributes $\frac{1}{2}h_4\gamma^4$, 3-space $\frac{3}{2}a_4'\gamma^4$, the extra times $\frac{3}{2}h_4\gamma^4 + \frac{3}{2}(H/f)\gamma^0$, with $h_4 = b_4'/(2b_4) - a_4'$.

The good sector: no dependence on x_5, x_6, x_7 (zero extra-time momenta), a coordinate 3-torus of side ℓ with the momenta $k \in \Delta k\, \mathbb{Z}^3$, $\Delta k = 2\pi/\ell$, and the extra-time coordinate volume v_t; the proper cell volume is $\ell^3 v_t e^{-a_{4,0}} b_{4,0}^2$. With the fold conditions (ASSUMED filling of both) and the chosen tip condition (section 5; together they kill the y-current and make the blocks h_j self-adjoint) the good sector is quantised in the positive realisation of the canonical quantisation of `Revision/SPEC.md` section 6; the field-theory record `Revision/docs/DIRAC16COMPLEX_FIELD_THEORY` (its section 11) constructs that realisation explicitly only for single good-sector momenta with frozen coefficients, and without a boundary condition at the degenerate points the curved good sector has complex frequencies wherever $m^2 < \frac{9}{4}H^2 e^{2a_4}/b_4$. The expectation-value rule $\langle \Psi^\dagger M \Psi \rangle = \mathrm{Tr}(M\rho)$, $\rho = \sum_{\mathrm{occ}} f\, uu^\dagger B$, defines the quasi-free (Slater or Mermin) states used here.

item	**Wolfram records**	**sympy records**
gammas, C, B, Γ	`fixture_input`, `clifford_relation`, `C_B_Gamma`	`fixture_input`, `fixture_python_equals_wolfram`, `clifford_relation`, `C_B_Gamma`, `BC_and_Cg4`
hidden coordinate, line element, $\sqrt{\lvert g\rvert}$	`geometry_hidden_coordinate`, `geometry_line_element`, `geometry_sqrt_det`	`geometry_hidden_coordinate`, `geometry_sqrt_det`, `geometry_line_element`
spin connection	`spin_connection_compatibility`, `spin_connection_slash`, `spin_connection_time_terms`, `spin_connection_slash_x0_chart`	`spin_connection_compatibility`, `spin_connection_slash`, `spin_connection_time_terms`, `spin_connection_slash_x0_chart`

2.1 The corrected metric in the Kohn-Sham record

On 2026-10-16 the primordial metric was corrected (version 2; the hidden direction is space-like and carries $e^{-2a_4} b_4 \cot^2 z$, 3-space carries e^{2a_4} without a warp, the extra times $-e^{-2a_4} b_4 (\sin^2 z)^{1/6}$, z real); the metric of 2026-10-14, for which the earlier versions of this record were computed, is withdrawn. The Kohn-Sham record was recomputed for it (phase W5 of `HANDOFF.md`, 2026-10-16/17):

1. The reduction (records `ansatz_removes_spin_connection`, `ansatz_without_y_factor_term_survives`, `ansatz_without_x4_factor_term_survives`, `ansatz_norm`): the factor $e^{-(3/2)Hy} e^{a_4/2} b_4^{-1}$ of the orbitals is the half-density factor of $\sqrt{|g|}$; it removes both spin-connection terms, and the Hilbert norm of the reduced problem is $\int \chi^\dagger \chi \, dy$ for every a_4 and b_4. The hidden direction enters the block Hamiltonians through $1/f$, the 3-momenta through $\kappa = e^{-a_4}$, which does not depend on y (before: e^{-Hy-a_4}).

2. The densities (records `ks_coupling_volume_scaling`, `history_solver_units`): a proper density is the orbital sum times $P = e^{-3Hy}/\mathrm{Vol}_7$ with $\mathrm{Vol}_7(a_{4,0}) = \ell^3 v_t\, e^{(2\beta-1)a_{4,0}} =$

$\ell^3 v_t e^{(5/3)a_{4,0}}$ (the proper volume in units of $b_4(0)^2$); the solvers use the coupling $\hat{\lambda} = \lambda\, b_4(0)^{-2}$. A Kohn-Sham state depends on the coupling and the cell volume only through their ratio: the slice state at $(\hat{\lambda}, \mathrm{Vol}_7(a_{4,0}))$ is the state of the coordinate cell $\ell^3 v_t$ at the effective coupling $\hat{\lambda}\, e^{-(5/3)a_{4,0}}$, at the same f and κ.

3. The boundary conditions: the fold joins the two halves of a cell, a spinor continuous across it has the same $\chi(y)$ on both halves, and the two fold conditions $\chi_2(0) = 0$ and $\chi_1(0) = 0$ are both solved and filled (ASSUMED; section 5). The construction of the withdrawn metric, a Z2 orbifold at the patch end whose mirror copy carried $-m$, does not exist in the corrected metric.

4. The energy along the history (record `emt_energy_change_history`): $dE/da_4 = -3P_3 - 3(\beta/2-1)P_t - (\beta/2-1)P_0 = -3P_3 + P_t + \frac{1}{3}P_0$ with $P = 2\mathrm{Vol}_7(a_{4,0}) \int e^{3Hy} p\, dy$ (section 9).

5. The parameters were re-derived from the stated rules (section 5): $N = 8, 592, 2352$ and the couplings of the table there; every state of the matrix changed.

3 The exact block reduction

Ansatz. $\Psi = e^{ik\cdot x} e^{-(3/2)Hy} e^{a_4/2} b_4^{-1} \chi(y, x_4)$ gives EXACTLY, for a time-dependent a_4 and b_4,

$$\tfrac{1}{f}\gamma^0 \partial_y \chi + \gamma^4 \partial_{x_4}\chi + i\kappa\, k\cdot\gamma\, \chi = (M_{\mathrm{eff}} - i v_v \gamma^4)\chi, \qquad \kappa(x_4) = e^{-a_4(x_4)}, \qquad f(x_4) = e^{-a_4} b_4^{1/2},$$

i.e. $i\partial_{x_4}\chi = h(x_4)\chi$ with h Hermitian in $\int dy\, \chi^\dagger \chi$. The factor $e^{-(3/2)Hy}$ removes the term $\frac{3}{2}(H/f)\gamma^0$ and the factor $e^{a_4/2} b_4^{-1}$ the term $(b_4'/b_4 - \frac{1}{2}a_4')\gamma^4$; without either the term survives. At a fixed instant the instantaneous (adiabatic) problem is $h(a_{4,0}, f_0)\chi = \varepsilon\chi$ with $a_{4,0} = a_4(x_4)$ and $f_0 = f(x_4)$.

Blocks. $J = \gamma^0\gamma^1\gamma^4$ ($J^2 = 1$), $K_1 = \gamma^2\gamma^3$ and $K_2 = \gamma^5\gamma^6$ ($K^2 = -1$) commute with each other and with γ^0, $\gamma^0\gamma^1$, $\gamma^0\gamma^4$, B and C; their joint eigenspaces, labelled (j, s_2, s_3) with $j, s_2, s_3 = \pm 1$, are eight rank-2 blocks with an explicit unitary basis V (entries of $2\sqrt{2}\,V$ in $\{0, \pm 1, \pm i\}$). For k along x_1 (the spinor rotation commutes with γ^0, γ^4, B, C, so the spectra depend on $|k|$ only), every block of type j carries

$$h_j = j\big[-\tfrac{i}{f}\sigma_1 \tfrac{d}{dy} + M_{\mathrm{eff}}(y)\sigma_2 + \kappa k \sigma_3\big] + v_v(y),$$

equivalently $\chi' = fN\chi$ with $N = M_{\mathrm{eff}}\sigma_3 - \kappa k \sigma_2 + ij(\varepsilon - v_v)\sigma_1$; in the real form $\chi = (a, ib)$, $a' = f[M_{\mathrm{eff}} a - (\kappa k + j(\varepsilon - v_v))b]$ and $b' = f[(j(\varepsilon - v_v) - \kappa k)a - M_{\mathrm{eff}} b]$. In the block basis $\gamma^0 = \sigma_3$, $B = j s_2$ and $C = s_2\sigma_2$.

Symmetries and degeneracies. $h_{-1} - v = -(h_{+1} - v)$ and $\sigma_3 h_j(k)\sigma_3 = h_{-j}(-k)$: the (j, k) and $(-j, -k)$ orbitals are degenerate. On a lattice shell $|k|^2 = \Delta k^2 n_2$ with $r_3(n_2)$ vectors every level of $h_j(|k|)$ is $4r_3(n_2)$-fold for each j; at $k = 0$ every level is 4-fold per j. The chirality Γ maps block (j, s_2, s_3) onto $(-j, s_2, s_3)$ with the 2×2 entry $s_2\sigma_2$, and $\sigma_2 h_j(M, k)\sigma_2 = h_{-j}(-M, k)$ with the two fold conditions exchanged (the ingredient of T3, section 12).

Where the deflation enters (exact). $a_{4,0}$ and $b_{4,0}$ enter h_j through $\kappa k = k e^{-a_{4,0}}$ (the 3-momenta redshift as $k e^{-a_4}$) and through the hidden scale $f_0 = e^{-a_{4,0}} b_{4,0}^{1/2}$, which multiplies the hidden

derivative by $1/f_0$ (the proper hidden length of the cut-off cell is $f_0 L$); because the good sector has no extra-time momentum, the extra times themselves enter only through the proper cell volume, that is through the densities and the mean field (sections 2.1 and 10).

item	**Wolfram records**	**sympy records**
ansatz and Hamiltonian	`ansatz_removes_spin_connection`, `ansatz_without_y_factor_term_survives`, `ansatz_without_x4_factor_term_survives`, `ansatz_norm`, `hamiltonian_16_hermitian`	`ansatz_removes_spin_connection`, `ansatz_without_y_factor_term_survives`, `ansatz_without_x4_factor_term_survives`, `ansatz_norm`, `hamiltonian_16_hermitian`
blocks	`blocks_commuting_set`, `blocks_basis_unitary`, `blocks_forms`, `blocks_relation_to_Gamma`, `block_hamiltonian`, `block_ode_equivalent`	`blocks_commuting_set`, `blocks_projectors`, `blocks_basis_unitary`, `blocks_forms`, `blocks_not_everything_block_diagonal`, `blocks_relation_to_Gamma`, `block_hamiltonian`, `block_ode_equivalent`
symmetries	`block_type_relation`, `block_Gamma_map`, `rotation_invariance`	`block_type_relation`, `block_Gamma_map`, `rotation_invariance`
solver (numerical basis)		`theory_input_metric_forms`, `block_reduction_numeric`, `gamma_fixture_numeric` (Rust solver report)

4 The functional: Hartree, exact uniform-gas exchange, Mermin

Exchange (exact for the uniform gas). The interaction is $U = \frac{\lambda}{2} S^2$, $S = \bar{\Psi}\Psi$, normal ordered ($\lambda > 0$ repulsive). For a quasi-free state, Wick's theorem gives $\langle : S^2 : \rangle = (\operatorname{Tr} C\rho)^2 - \operatorname{Tr}(C\rho C\rho)$. For every $p \to -p$ symmetric occupation of the 8-fold good-sector gas, at every temperature, $\rho = (nB + SC)/16$ and

$$e_H = \tfrac{\lambda}{2} S^2, \qquad e_x = -\tfrac{\lambda}{32}(n^2 + S^2), \qquad e_{\mathrm{int}} = \tfrac{15}{32}\lambda S^2 - \tfrac{1}{32}\lambda n^2,$$

$$M_{\mathrm{eff}}(y) = m + \tfrac{15}{16}\lambda S(y), \qquad v_v(y) = -\tfrac{\lambda}{16} n(y),$$

with proper densities; for one filled level at rest $E_x/E_H = -1/8$, and on shell $\langle \mathcal{L} \rangle / \sqrt{|g|} = e_{\mathrm{int}}$. In the solvers' units (section 2.1) the same formulas hold with $\lambda \to \hat{\lambda}$ and every density per $\mathrm{Vol}_7(a_{4,0})$. No correlation is included (Hartree plus exchange only). The Rust solver reads the coefficients from `ks-theory.json` and checks them (`theory_input_coefficients`: $M_{\mathrm{eff}} = m + 0.9375\lambda S$, $v_v = -0.0625\lambda n$).

Exact Fock exchange of the slab (diagnostic and variant). For closed shells the exact Fock exchange of the Kohn-Sham determinant is also local, $e_x^{\text{exact}} = -\frac{\lambda}{32}(n^2 + S^2 - Q^2 - Y^2)$ with $Q = \langle \Psi^\dagger B \gamma^0 \Psi \rangle$ and $Y = 0$; the canonical functional is the uniform-gas one, the difference $\Delta E_x = \frac{\lambda}{32} 2\text{Vol}_7 \int e^{3Hy} Q^2 \, dy$ is reported for every state, and the exact-Fock functional is solved as a variant (60 states, `Revision/kohn_sham/results/exx/exact-fock-variant.csv`).

Densities, energy, Mermin. Per orbital, with $P(y) = e^{-3Hy}/\text{Vol}_7(a_{4,0})$: $n_o = P\chi^\dagger\chi$, $s_o = P\chi^\dagger j\sigma_2\chi$, $t_o = P\chi^\dagger j\sigma_3\chi$, $q_o = P\chi^\dagger\sigma_3\chi$. Totals are $\sum w\, g\, f$ times these, $g = 4r_3(n_2)$, $w = 1/2$ (section 5). The Kohn-Sham energy of the doubled system is $E_{KS} = \sum gf\varepsilon - 2\text{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{\text{int}} \, dy$; Mermin's occupations are $f = 1/(1 + e^{(\varepsilon-\mu)/T})$ with μ fixed by $\sum gf = N$, the entropy is $S_{\text{ent}} = -\sum g[f \ln f + (1-f)\ln(1-f)]$, the grand potential (the thermodynamic effective potential) is $\Omega = -T\sum g \ln(1 + e^{-(\varepsilon-\mu)/T}) - 2\text{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{\text{int}} \, dy$, and $F = \Omega + \mu N$.

Discrepancy recorded, not hidden: the line "mu fixed by sum w g f = N" of `ks-theory.json` is inconsistent with its own densities, energy and grand potential; the solver and the reference use $\sum gf = N$, and `Revision/kohn_sham/results/parameters.json` records the discrepancy.

item	Wolfram records	sympy records
uniform gas and exchange	`gas_mode_projector`, `gas_angular_average`, `gas_densities`, `exchange_uniform_gas`	`gas_mode_projector`, `gas_angular_average`, `gas_negative_energy_modes`, `gas_densities`, `hf_wick_contraction`, `exchange_uniform_gas`, `filled_shell_ratio`
Kohn-Sham potentials	`ks_potentials`, `ks_onshell_lagrangian`	`ks_potentials`, `ks_onshell_lagrangian`, `ks_theory_json_exchange`
coupling, cell volume and history	`ks_coupling_volume_scaling`, `history_solver_units`	`ks_coupling_volume_scaling`, `history_solver_units`
exact Fock exchange of the slab	`exchange_slab_exact_fock`	`exchange_slab_exact_fock`

5 Boundary conditions, filling convention, numerics and parameters

Fold (both fold conditions filled, ASSUMED). The fold $\cos z = 0$ at $y = 0$ joins the two halves of a cell; the reduced equation is the same on both halves ($dy = \cot z \, dx_0$, $e^{x_0} dx_0 = f \, dy$, $\gamma^\mu \Omega_\mu$ free of x_0), so a spinor continuous across the fold has the same $\chi(y)$ on both halves; the boundary form of the doubled cell at the fold is $2(-ij/f)\phi^\dagger\sigma_1\chi(0)$, and self-adjointness needs $(1 - Q(\theta))\chi(0) = 0$. The record fills both fold conditions, the even $\theta = 0$ ($\chi_2(0) = 0$, $b(0) = 0$) and the odd $\theta = \pi$ ($\chi_1(0) = 0$, $a(0) = 0$), each normalised on one half; N is the particle number of the whole cell, each half holds $N/2$, and $w = 1/2$ per half. That both are filled is ASSUMED (the record's convention); there is no mirror copy.

Tip (chosen). At $y = -L$: $(1 - Q(\theta))\chi(-L) = 0$, $Q(\theta) = \cos\theta\,\sigma_3 + \sin\theta\,\sigma_2$, canonical $\theta = 0$, cutoff $L = 3$. With κ independent of y nothing suppresses an orbital toward the tip: for constant M and v

the branches are $e^{\pm f\lambda y}$, $\lambda = \sqrt{M^2 + \kappa^2 k^2 - (\varepsilon - v)^2}$. The dependence on L is measured in `Revision/kohn_sham/tip_convergence/` ($L = 3$ to 6 in steps of 0.5 with the record's step $h = 1/300$, every run repeated at $h/2$; a copy of the solver that reads L from the environment reproduces the committed quick matrix byte for byte at $L = 3$; the reference solver agrees at $L = 3$ and 4; report `Revision/kohn_sham/tip_convergence/tip-convergence.json`, 7 of 7 checks PASS; rerun for the corrected metric on 2026-10-17). The free states are exact fold-band gases and do not depend on L: E_{KS}, HOMO, LUMO and the gap of every free state with $N = 8$ and 592 and of $N = 2352$ at $a_{4,0} = 1, 2$ are the same at every L; only the values at the fold change, through the normalisation of the fold-band orbital, $\rho(0; L) = \rho(0; \infty)/(1 - e^{-2fmL})$ (the recorded $L = 3$ fold values are high by 0.25%, 1.38% and 4.81% at $a_{4,0} = 0, 1, 2$). The $k = 0$ bulk levels approach m only algebraically, like the levels of a box of proper length fL: the bulk edge 1.2923 that defines $N = 2352$ is an $L = 3$ value (1.0977 at $L = 6$), and the free $N = 2352$, $a_{4,0} = 0$ state changes its occupied set from $L = 3.5$ on (its gap 0.01754 closes; $E_{KS} = 2288.388$ at $L = 3$ and 2245.486 at $L = 6$, not converged). The interacting zero modes depend on L, the more the later the slice: their proper density grows toward the tip like $e^{(3H-2fm)|y|}$, and for $N = 8$ the recorded E_{KS} is $\mp 3.7787 \times 10^{-2}$ against $\mp 3.8688 \times 10^{-2}$ at large L at $a_{4,0} = 0$, $\mp 1.4571 \times 10^{-2}$ against $\mp 2.1622 \times 10^{-2}$ at $a_{4,0} = 1$ and $\mp 6.980 \times 10^{-3}$ against $\mp 3.239 \times 10^{-2}$ at $a_{4,0} = 2$ ($\hat{\lambda} = \pm\lambda_1$; 0.977, 0.674 and 0.215 of the large-L values); the interaction shifts $E_{KS}(\pm\lambda_1) - E_{KS}(0)$ of $N = 592$ and 2352 at $a_{4,0} = 2$ are 4.6 to 4.7 times larger at large L than at $L = 3$. 17 of the 19 interacting states studied converge (geometric tails); the self-consistent iteration fails only for $N = 2352$ at $a_{4,0} = 0$, from $L = 3.5$ for $+\lambda_1$ and from $L = 4.5$ for $-\lambda_1$ (a failure of the iteration, not a proof that no self-consistent state exists). The record's calibration rule applied at each L drives λ_1 to zero: 157.7, 2.131 and 0.5363 at $L = 3$, 1.249, 0.01687 and 1.03×10^{-5} at $L = 6$ ($N = 8, 592, 2352$).

Exact $k = 0$ spectra. For constant $M > 0$, $v = 0$, $\theta = 0$: the even fold condition has $\varepsilon = 0$ with the fold zero mode $\chi = (e^{fMy}, 0)$, and $\varepsilon = \pm\sqrt{M^2 + (n\pi/(fL))^2}$; the odd fold condition has $\varepsilon = \pm\sqrt{M^2 + p^2}$ with $\tan(pfL) = -p/M$. The even $j = +1$ fold band is $\varepsilon = \kappa k$ EXACTLY for every k, with the orbital $(e^{fMy}, 0)$ (record `free_rescaling_relation_and_band_monotone`); its slope is $c = e^{-a_{4,0}}$ for every M, L and f (record `brane_band_slope`; the record keeps the name of the earlier construction).

Filling (CONVENTION, justification OPEN). Particles occupy the positive branch (the levels with $\varepsilon > 0$ of the $\lambda = 0$ problem) and the $k = 0$ fold zero modes, each followed continuously in λ; the negative branch is the normal-ordered sea and contributes nothing to N, n, S or the energy-momentum tensor. The excluded thermal sea holes are reported as a diagnostic (section 8).

Numerics. The Rust solver (`Revision/kohn_sham/solver`, own crate, no external crates) solves each block by shooting with RK4 on 900 steps, Pruefer counting and a root tolerance of 1e-13, Anderson-mixed SCF to 1e-11; when the direct iteration diverges it falls back to a continuation in the coupling ($\hat{\lambda} \times \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ with half the mixing). The independent Python reference (`Revision/kohn_sham/reference`) uses a staggered finite-difference discretisation on three grids (300, 600, 1200) with Richardson extrapolation and a stated uncertainty, validated on a fourth grid, with its own continuation fallback; its convergence order is two (record `free_convergence_order_two`).

Parameters (`Revision/kohn_sham/results/parameters.json`): $H = m = 1$, $L = 3$, $\Delta k = 0.25$, $v_t = 1$, $\mathrm{Vol}_7(0) = 15875.21366031351$ and $\mathrm{Vol}_7(a_{4,0}) = \mathrm{Vol}_7(0)\, e^{(5/3)a_{4,0}}$, tip $\theta = 0$, history $A = 1$, $\beta = 4/3$, slices $a_{4,0} \in \{0, 0.5, 1, 1.5, 2\}$, temperatures $T \in \{0.01, 0.02, 0.05\}$. Particle numbers: $N = 8$ (the $k = 0$ fold zero modes of both block types), $N = 2352$ (the largest closed shell of the free $a_{4,0} = 0$ aufbau whose last level lies below the bulk edge 1.292292828068783, the lowest level not

on the even $j = +1$ fold band) and $N = 592$ (the closed shell nearest 2352/4). Couplings: λ_1 and λ_2 (values of $\hat{\lambda}$) keep the first-order mean-field potential below 0.1 and 0.3 (units of m) along the whole history; the strength is the largest first-order potential per unit $\hat{\lambda}$ over the five slices, with the densities per $\mathrm{Vol}_7(a_{4,0})$:

N	**strength per unit lambda_hat**	**lambda_1**	**lambda_2**
8	0.0006341788	157.7	473.1
592	0.04692923	2.131	6.393
2352	0.1864486	0.5363	1.609

The canonical matrix holds 75 ground states (3 values of N, the five couplings $0, \pm\lambda_1, \pm\lambda_2$, five slices), each with its first excited state, its a_4 neighbours, its rescaling partner and, for $\lambda \neq 0$, its exact-Fock variant, and 135 Mermin states ($\hat{\lambda} \in \{0, \pm\lambda_1\}$, three temperatures).

item	**records**
boundary conditions (Wolfram and sympy)	`bc_fold_same_equation_on_both_ha lves`, `bc_fold_conditions`, `bc_self_ adjoint_boundary_term`, `bc_tip_family`, `bc_current_conserved_along_y`, `bc_exact_k0_spectra`, `bc_tip_a symptotics`, `brane_band_slope`
free problem (Rust solver)	`free_k0_analytic_spectra`, `free_zer o_mode_exact`, `free_brane_band_slope`, `free_block_type_symmetries`, `free_par ticle_branch_labels`, `free_tip _angle_boundary_identity`
free problem and parameters (reference)	`free_k0_analytic_spectra`, `free_con vergence_order_two`, `free_k0_block_ty pe_symmetry`, `free_zero_mode`, `free_bra ne_band_slope`, `free_particle_branch`, `parameters_closed_shells`, `paramete rs_calibration`, `richardson_asymp totic_ratio`, `richardson_uncer tainty_validated`
parameters (cross-check)	`problem_definition_identical`, `parameters_particle_numbers`, `parameters_couplings`

6 Instantaneous ground and first excited states along the prescribed history

The history is $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$ with $A = 1$ (the exponentially deflating member: the scale factor of the extra times is $b_4(0)^{1/2}e^{-Hx_4/3}(\sin^2 z)^{1/12}$), PRESCRIBED, not solved for (section 11). At every slice $a_{4,0}$ the ground state is the self-consistent aufbau state of the instantaneous problem; the first excited state is computed three ways: the Kohn-Sham gap $\varepsilon_{\mathrm{LUMO}} - \varepsilon_{\mathrm{HOMO}}$; the ensemble Δ-SCF energy (one particle moved from the highest occupied degenerate group to

the lowest empty group, spread uniformly over each group, self-consistent); and the list of the 24 lowest particle-hole excitations (`Revision/kohn_sham/results/excited/particle-hole/`). Every ground state of the matrix is closed-shell.

Ground and first excited states for $N = 592$ (`Revision/kohn_sham/results/ground/summary.csv` and `Revision/kohn_sham/results/excited/summary.csv`):

lambda_hat	**a4,0**	**E_KS**	**KS gap**	**Delta-SCF**
0	0	361.1161	0.03858678	0.03858678
0	1	132.8472	0.01419528	0.01419528
0	2	48.87176	0.005222153	0.005222153
2.131	0	358.301	0.03858459	0.03858459
2.131	1	131.7652	0.01419487	0.01419487
2.131	2	48.35502	0.005222125	0.005222125
-2.131	0	363.8955	0.03858511	0.03858511
-2.131	1	133.9218	0.01419489	0.01419489
-2.131	2	49.38796	0.005222125	0.005222125
6.393	0	352.798	0.03857391	0.03857391
6.393	1	129.6573	0.01419216	0.01419216
6.393	2	47.3264	0.005221902	0.005221903
-6.393	0	369.2418	0.03857847	0.03857847
-6.393	1	135.982	0.0141925	0.0141925
-6.393	2	50.41245	0.005221906	0.005221906

The free Kohn-Sham gap along the history ($\lambda = 0$, `ground/summary.csv`):

N	**a4,0 = 0**	**a4,0 = 0.5**	**a4,0 = 1**	**a4,0 = 1.5**	**a4,0 = 2**
8	0.25	0.1516327	0.09196986	0.05578254	0.03383382
592	0.03858678	0.02340407	0.01419528	0.008609875	0.005222153
2352	0.01753795	0.01472852	0.0089333	0.00541832	0.003286377

Reading (labelled interpretation of these numbers): the free ground states occupy the fold band $\varepsilon = \kappa|k|$ and the $k = 0$ zero modes, so as the extra times deflate the 3-momenta redshift as $ke^{-a_{4,0}}$ and the occupied free levels redshift with them: the free energies of $N = 592$ and 2352 fall exactly like $e^{-a_{4,0}}$, the Kohn-Sham gap of $N = 8$ is $\kappa\Delta k = 0.25\, e^{-a_{4,0}}$ and that of $N = 592$ falls like $e^{-a_{4,0}}$; for $N = 2352$ the lowest empty level at $a_{4,0} = 0$ is the bulk edge (by the choice of N), and from $a_{4,0} = 0.5$ on, where the $k = 0$ bulk levels have risen above it, the next fold-band shell, so its gap falls like $e^{-a_{4,0}}$ from there; with interaction the energy changes also with the effective coupling $\hat{\lambda}e^{-(5/3)a_{4,0}}$. With $\lambda = 0$ the Δ-SCF energy equals the Kohn-Sham gap (worst difference 3.086e-13 over 15 cases, record `excited_delta_scf_free_equals_gap`); with interaction the two differ slightly, because the Δ-SCF state relaxes the mean field self-consistently.

item	Rust solver records	reference records	cross-check records
ground states	`ground_runs_completed`, `ground_scf_converged`, `ground_closed_shell`, `ground_window_complete`, `ground_N_conservation`, `ground_energy_two_forms`, `ground_orbital_boundary_conditions`	`ground_scf_converged`, `ground_a4_neighbours_tight`, `ground_grid_consistency`, `ground_closed_shell`, `ground_window_complete`, `ground_N_conservation`, `ground_energy_two_forms`	`ground_occupations_and_groups`, `ground_energies`, `ground_homo_lumo_gap`, `ground_eigenvalues`, `ground_profiles`
first excited state	`excited_delta_scf_free_equals_gap`, `excited_scf_converged`	`excited_delta_scf_free_equals_gap`, `excited_particle_hole_lowest_is_gap`	`excited_delta_scf`, `excited_particle_hole_lists`
exact-Fock variant	`exx_variant_scf_converged`, `exx_variant_y_conservation`, `exx_variant_occupation_order`	`exx_scf_converged`, `exx_N_conservation`, `exx_energy_two_forms`, `exx_emt_energy_integral`, `exx_y_conservation_integrated`	`exchange_delta_E_x`, `exx_variant_scf`
SCF path (information)	`thermo_scf_path`	`scf_paths`	

7 Adiabaticity and Fermi-level crossings

Exactly (mean field), $i\partial_{x_4}\chi = h(a_4(x_4), f(x_4))\chi$ conserves k, the block and the fold condition; transitions occur only between levels of the same $(k, j, \text{block}, \text{fold condition})$. The measure along $a_4 = AHx_4$ is

$$Q_{nm} = AH\,\frac{|\langle n|\partial_a h|m\rangle|}{(\varepsilon_n - \varepsilon_m)^2} = AH\,\frac{|\langle n|\partial_a m\rangle|}{|\varepsilon_n - \varepsilon_m|},$$
$$\partial_a h_j = -jk\kappa\sigma_3 + (1-\tfrac{\beta}{2})\,j(-\tfrac{i}{f}\sigma_1\partial_y) + j(\partial_a M_{\text{eff}})\sigma_2 + \partial_a v_v,$$

for n occupied and m empty in the same sector (the second term, one third of the kinetic term, comes from the shrinking hidden direction); the state is adiabatic if $\max Q_{nm} \ll 1$, and Q^2_{nm} is the leading-order transition probability. The self-consistent derivatives come from the states at $a_{4,0} \pm \delta$, $\pm 2\delta$ ($\delta = 2\times 10^{-3}$, Richardson); they contain the change of the effective coupling $\hat{\lambda}/\mathrm{Vol}_7(a_{4,0})$ along the history. Hellmann-Feynman, $d\varepsilon_n/da = \langle n|\partial_a h|n\rangle$, is checked in 75 cases (worst 6.444e-11, record `adiabatic_hellmann_feynman`).

Results (`Revision/kohn_sham/results/adiabatic/adiabaticity.csv`): because the hidden direction shrinks, $Q_{\max}$ is not zero even for $N = 8$ at $\lambda = 0$ (state N8_lam0_a00: 0.06814928); over

the $N = 8$ states the largest value is $Q_{\max} \leq 0.07273792$ (state N8_lamp2_a00, pair `0:-1:even:0 -> 0:-1:even:1`). The largest value over the whole matrix is 0.1982205 (state N2352_lamm1_a00, pair `26:+1:even:0 -> 26:+1:even:1`: two levels of the same shell, block type and fold condition). For $\lambda = 0$:

N	**a4,0**	**Q_max**	**delta eps of the pair**	**dE/da4 (EMT)**	**dE/da4 (differences)**
592	0	0.1396883	0.8591647	-361.1161	-361.1161
592	0.5	0.09450268	1.181936	-219.028	-219.028
592	1	0.06819305	1.503745	-132.8472	-132.8472
592	1.5	0.05108696	1.826609	-80.5759	-80.5759
592	2	0.03879691	2.167139	-48.87176	-48.87176
2352	0	0.194329	0.6543959	-2288.388	-2288.388
2352	0.5	0.11849	0.9955121	-1387.978	-1387.978
2352	1	0.0782197	1.362942	-841.851	-841.851
2352	1.5	0.05520876	1.730963	-510.6084	-510.6084
2352	2	0.04047331	2.105635	-309.6997	-309.6997

The last two columns are the energy change along the history, from the energy-momentum tensor ($dE/da_4 = -3P_3 + P_t + \frac{1}{3}P_0$, $P = 2\mathrm{Vol}_7(a_{4,0}) \int e^{3Hy} p \, dy$, section 9) and from finite differences at fixed occupations; for the free fold-band gases both equal $-E_{KS}$.

Fermi-level crossings. When levels of different k cross the Fermi level between slices, the instantaneous aufbau ground state differs from the adiabatically continued state, and the solver flags it. No crossing occurs in the canonical matrix: every one of the 15 series of `Revision /kohn_sham/results/adiabatic/history.json` has an empty crossing list, and `adiabatic/fermi-level-crossings.csv` records no change of the occupied set. A demonstration at $\lambda = 0$ and $N = 2360$ (the smallest closed shell of the $a_{4,0} = 0$ aufbau that occupies a level off the even $j = +1$ fold band; `adiabatic/crossing-demo.csv`):

a4,0	**occupied set as at a4,0 = 0**	**open shell**	**E aufbau**	**E adiabatically continued**	**difference**
0	true	false	2298.727	2298.727	0
0.5	false	true	1394.281	1398.957	4.676354
1	false	true	845.6741	853.6174	7.943281
1.5	false	true	512.9273	523.3331	10.40583
2	false	true	311.1061	323.5841	12.47799

From $a_{4,0} = 0.5$ on, the fold-band levels redshift ($\varepsilon = e^{-a_4} k$) while the $k = 0$ bulk levels rise ($\varepsilon = \sqrt{M^2 + (n\pi/(fL))^2}$, $f = e^{-a_4/3}$): the instantaneous ground state is not the adiabatically reached state, which lies above it by the last column.

Interpretation (labelled): with $Q_{\max} \leq 0.1982205$ the instantaneous states are self-consistently adiabatic for the same-sector transitions at the rate $A = 1$ in the canonical matrix only in the weak sense $Q_{\max} < 1$; the largest values approach 0.2 for $N = 2352$ at $a_{4,0} = 0$. This does not solve the time-dependent problem (OPEN), and where Fermi-level crossings occur (the demonstration) the instantaneous ground state is not the state that the evolution reaches.

Rust solver records	reference records	cross-check records
adiabatic_hellma nn_feynman, adiabati c_crossing_flag_ demonstration	adiabatic_hellma nn_feynman_discrete, crossing_demo_reference	adiabatic_dE_da4, adiabatic_Q_max, crossing_demonstration

8 Mermin thermodynamics

The Mermin states at $T = 0.01, 0.02, 0.05$ solve the self-consistent problem at fixed N with μ the root of $\sum gf = N$ in a well-conditioned form (thermal particles above and holes below a split of the levels, `solver/src/mermin.rs`). The checks: Ω in its two forms, $-dF/dT = S_{\rm ent}$, $C_V = T\,dS_{\rm ent}/dT = dE/dT$ at fixed N, and the chemical potential against 40-digit roots on the solver's own final levels (40-digit and 50-digit roots agree to 1.5e-38, record `root_precision_s elf_check`). For $N = 592$, $\lambda = 0$ (`Revision/kohn_sham/results/thermo/thermodynamics.csv`):

a4,0	T	mu	E	entropy	Omega	sea holes / N
0	0.01	0.8099438	361.6116	76.01079	-118.6352	3.76e-48
0	0.02	0.8110736	362.7433	152.6427	-120.4652	3.73e-25
0	0.05	0.8078443	369.9279	358.7396	-126.2529	3.43e-11
2	0.01	0.1078013	51.39934	517.9789	-17.59884	5e-08
2	0.02	0.09893307	58.47195	991.924	-19.93491	0.000221
2	0.05	0.03959481	96.58455	2112.111	-32.46113	0.234

Validity of the particle-only ensemble (diagnostic of the CONVENTION). For every thermal state the number of thermal holes that the excluded sea fold band would carry at the same μ and T is computed. It exceeds 1% of N in 27 of 135 states, most of them at $N = 8$ and all at $a_{4,0} \geq 0.5$; the largest is 594.496 N (record `thermo_sea_hole_diagnostic_computed`). There the particle-only Mermin ensemble is outside its range of validity and a thermal treatment of the sea (pairs) would be required; in the table above this is the row $a_{4,0} = 2$, $T = 0.05$ (state N592_lam0_a20_T50).

Rust solver records	reference records	cross-check records
thermo_runs_completed, thermo_scf_converged, thermo_N_conservation, thermo_grand_pot ential_two_forms, thermo_entropy_identity, thermo_CV_identity, thermo_window_cut, thermo_window_sh ells_beyond, thermo_m u_well_conditioned_root, thermo_mu_vs_40d igit_roots, thermo_s ea_hole_diagnost ic_computed, thermo_s cf_path	thermo_scf_converged, thermo_N_conservation, thermo_grand_pot ential_two_forms, thermo_CV_identity, thermo_entropy_identity, thermo_window_cut, thermo_sea_holes_tail, scf_paths	thermo_state_functions, thermo_derivatives, thermo_sea_hole_ diagnostic, thermo_l evels, thermo_mu_high_p recision, thermo_m u_rounding_diagnostic, thermo_mu_rust_s tated_bound

Mermin-root report	records
Revision/kohn_sham/reports/ks-ru st-mermin-roots.json	solver_runs_completed, root_pre cision_self_check, solver_mu_within _rounding_bound, single_reproduce s_committed_matrix, fixture_written

9 Energy-momentum tensor profiles

With the sign convention of Revision/SPEC.md section 4 ($\rho = -T^{x_4}{}_{x_4}$, $p_\mu = T^{\mu}{}_{\mu}$) and $T^{\mu}{}_{\nu} = -\langle K^{\mu}{}_{\nu}\rangle + \delta^{\mu}_{\nu} e_{\mathrm{int}}$, the proper profiles are

$$\rho(y) = \sum wgf\, \varepsilon\, n_o - e_{\mathrm{int}}, \qquad p_3(y) = \sum wgf\, \tfrac{\kappa|k|}{3}\, t_o + e_{\mathrm{int}}, \qquad p_t(y) = e_{\mathrm{int}},$$
$$p_0(y) = \sum wgf\, \left[(\varepsilon - v_v) n_o - M_{\mathrm{eff}}\, s_o - \kappa|k|\, t_o\right] + e_{\mathrm{int}},$$

with $T^{x_4}{}_{y} = 0$ for eigen-orbitals. The extra-time pressure p_t has no kinetic part (good sector). Exact identities, checked for every state: the trace of the kinetic parts is $M_{\mathrm{eff}} S + v_v n$; conservation along the hidden direction, $p_0' + 3H(p_0 - p_t) = 0$; the energy change $\nabla_\mu T^{\mu}{}_{x_4} = -\partial_{x_4}\rho - 3a_4'(\rho + p_3) - 3h_4(\rho + p_t) - h_4(\rho + p_0)$, i.e. on the history $dE/da_4 = -3P_3 + P_t + \frac{1}{3}P_0$ with $P = 2\mathrm{Vol}_7(a_{4,0}) \int_{-L}^{0} e^{3Hy} p\, dy$; and $2\mathrm{Vol}_7 \int e^{3Hy} \rho\, dy = E_{KS}$. Integrated profiles ($2\mathrm{Vol}_7(a_{4,0}) \int_{-L}^{0} e^{3Hy} X\, dy$, Revision/kohn_sham/results/ground/emt-integrals.csv), $\lambda = 0$:

N	a4,0	int rho	int p3	int p__t	int p0
592	0	361.1161	120.372	0	0
592	0.5	219.028	73.00934	0	-9.693924e-17
592	1	132.8472	44.2824	0	-6.658418e-16
592	1.5	80.5759	26.85863	0	-1.115453e-15
592	2	48.87176	16.29059	0	0
2352	0	2288.388	762.7961	0	0

N	a4,0	int rho	int p3	int p__t	int p0
2352	0.5	1387.978	462.6592	0	3.00941e-14
2352	1	841.851	280.617	0	-7.850057e-16
2352	1.5	510.6084	170.2028	0	-6.395636e-17
2352	2	309.6997	103.2332	0	4.739727e-16

For a fold-band orbital $(e^{fMy}, 0)$ with $\varepsilon = \kappa|k|$ the hidden-direction kinetic pressure vanishes identically, so the free states have $p_0 = p_t = 0$ (the table shows rounding values) and $\int p_3 = E_{KS}/3$: a radiation-like gas of the 3-space momenta, whose energy falls like e^{-a_4} ($dE/da_4 = -E_{KS}$, section 7). The profiles on 151 points $y = -3 + 0.02i$ are in `Revision/kohn_sham/results/ground/profiles/` (one file per state; densities per $\mathrm{Vol}_7(a_{4,0})$, that is the proper densities times $b_4(0)^2$). The energy density varies along the hidden direction in every state with a nonzero tensor (section 11). Equations of state seen by a 3-space observer are not part of this document; they are the work of `Revision/dark_sector/dirac16complex` (its Kohn-Sham history report `Revision/dark_sector/dirac16complex/reports/ks-history-run.json`: 8 of 8 PASS).

item	Wolfram and sympy records	Rust solver records	reference and cross-check records
tensor and identities	`emt_orbital_components`, `emt_trace_identity`, `emt_y_conservation_orbital`, `emt_y_conservation_selfconsistent`, `emt_x4_component`, `emt_energy_change_history`	`emt_energy_integral`, `emt_trace_derivative_form`, `emt_y_conservation_integrated`, `emt_y_conservation_pointwise`, `emt_energy_change_dE_da4`	`emt_energy_integral`, `emt_y_conservation_integrated`, `emt_energy_change_dE_da4`, `emt_integrals`, `emt_brane_tip_values`

10 The rescaling partners

Exact identity. With the proper hidden coordinate $Y = f_0 y$, $f_0 = e^{-a_{4,0}} b_{4,0}^{1/2}$,

$$\mathrm{KS}(a_{4,0}, b_{4,0};\, H, L, \Delta k, v_t, \lambda, N, T) = \mathrm{KS}\big(0, 1;\, H/f_0,\, f_0 L,\, \Delta k\, e^{-a_{4,0}},\, v_t\, e^{-3a_{4,0}} b_{4,0}^{3/2},\, \lambda, N, T\big)$$

for the levels, orbitals ($\chi_Y = \chi/\sqrt{f_0}$), proper densities and energy-momentum profiles; on the history ($b_{4,0} = e^{(4/3)a_{4,0}}$, $f_0 = e^{-a_{4,0}/3}$) the partner has $He^{a_{4,0}/3}$, $Le^{-a_{4,0}/3}$, $\Delta k\, e^{-a_{4,0}}$ and $v_t\, e^{-a_{4,0}}$ at the same $\hat{\lambda}$. Reading (labelled): the slice $a_{4,0}$ of the deflating history is the slice 0 with a 3-torus larger by $e^{a_{4,0}}$ in each direction, a shorter and more strongly warped hidden direction and a smaller extra-time coordinate volume: the 3-momenta redshift while the hidden direction and the extra times shrink.

Computed. Each solver solves the partner independently (60 partners). The worst level difference is 1.643e-13 and the worst relative difference of E_{KS} and of the profiles 8.193e-13 (records `rescalin`

g_identity_between_slices and rescaling_identity_energy_profiles). For $N = 592$, $\hat{\lambda} = \lambda_1$ (Revision/kohn_sham/results/rescaling/rescaling.csv):

a4,0	partner Delta k	partner v_t	partner H	partner L	max delta eps	relative delta E	partner E_KS
0.5	0.1516327	0.6065307	1.18136	2.539445	1.008e-13	1.307e-15	217.3785
1	0.09196986	0.3678794	1.395612	2.149594	8.704e-14	3.451e-14	131.7652
1.5	0.05578254	0.2231302	1.648721	1.819592	9.859e-14	3.952e-14	79.82634
2	0.03383382	0.1353353	1.947734	1.540251	9.148e-14	2.448e-13	48.35502

Wolfram and sympy records	Rust solver records	reference and cross-check records
rescaling_identity	free_rescaling_relation_and_band_monotone, rescaling_identity_between_slices, rescaling_identity_energy_profiles	rescaling_identity_reference, rescaling_partners

11 The recorded states are not admissible sources of the a_4 and b_4 equations

Revision/SPEC.md section 7 names the Kohn-Sham energy-momentum tensor as the source of the field equations for a_4 and b_4. The Einstein-Lovelock equations of the corrected metric (Revision/field_equations_a4, key generalSource of a4-equations.json) require of any source: (C1) every component independent of x_0 (the left-hand sides contain no x_0 for every a_4, b_4); (C2) $p_0 = p_t$ (the hidden and the extra-time components of every left-hand side are equal); (C3) no off-diagonal component, $T^{x_4}{}_{x_0} = 0$; (C4) on the history $a_4 = Hx_4$, $b_4 = e^{(4/3)Hx_4}$ with the author's weights and $\Lambda = -10$, the source $\kappa T - \Lambda$ that a4-equations.json records for this history (key exponentialHistory, author's weights), a function of $k = e^{(2/3)Hx_4}$ with $p_0 = p_t$. The record Revision/field_equations_a4/reports/ks-source-conditions.json (5 of 5 PASS) finds:

1. Every one of the 70 nonzero ground-state profiles depends on x_0: $(\max\rho - \min\rho)/\max|T| \geq 0.456213$ (record ks_profiles_depend_on_x0); C1 fails for every $a_4(x_4)$ and $b_4(x_4)$.
2. C2 holds to rounding for exactly the 10 nonzero states with $\lambda = 0$ (their fold-band orbitals have no hidden kinetic pressure and $e_{\text{int}} = 0$) and fails for every one of the 60 states with $\lambda \neq 0$ ($\max_y |p_0 - p_t|/\max|T| \geq 0.080929$; record ks_hidden_pressure_condition): the interaction bends the orbitals.
3. C3 holds: every Kohn-Sham eigen-orbital has $T^{x_4}{}_y = 0$ (record ks_offdiagonal_condition).
4. The history needs the source of C4, whose energy density changes sign along it ($\kappa\rho = 3.57407$ at $a_{4,0} = 0$ and -7.63383 at $a_{4,0} = 2$; its ratios do not depend on the unknown κ); the Kohn-Sham sources, averaged over x_0 per proper volume, have ratios $(p_3/\rho, p_0/\rho, p_t/\rho)$ that

differ from the needed ones by at least 1.3733 at every recorded state with a nonzero energy density (the free gas: $p_3/\rho = 1/3$, $p_0 = p_t = 0$): the history is a PRESCRIBED background (record `ks_history_is_a_prescribed_background`).

5. Five states have an identically vanishing tensor (the free $N = 8$ zero modes), a zero source, which is not admissible either: no exponential history is a vacuum solution for the author's weights (record `ks_zero_source_states_listed`).

The full analysis `Revision/field_equations_a4/ks_source/reports/ks-source-a4.json` (19 of 19 PASS) refines this exactly: the recorded states SATISFY the x_0 conservation identity $p_{0,y} + 3H(p_0 - p_t) = 0$ (largest residual 1.21233e-05 of the largest term on the output grid), so they fail the field equations because their components depend on x_0 (μ_1 of the energy density between 0.147738 and 0.738255 in the L1 measure) and, for $\lambda \neq 0$, because $p_0 \neq p_t$ even after averaging over x_0 (relative violation ≥ 0.00613418), not because conservation fails. The free gas is an exact fold-band gas: averaged over x_0 per proper volume it is $\bar{\rho} = Cb_4^{-2}$ with $p_3 = \rho/3$ and $p_0 = p_t = 0$, a source that conserves energy for every a_4 and b_4. With the author's weights the record's history needs $\kappa\rho = 3.57407$ at $a_4 = 0$ and -7.63383 at $a_4 = 2$, while the averaged free gas $N = 2352$ has $\bar{\rho} = 0.216249$ and 0.00104404 there: κ would have to run from 16.5276 to -7311.81, so no constant κ makes the gas the source. In a stated approximation (the x_0-averaged equations with the averaged free gas as the source, started on the record's history) the solution does not follow the history: with the author's weights the second-order system becomes singular at $x_4 = 0.428813$ ($a_4 = 0.361572$, $b_4 = 0.862627$, $a_4'/H = 0.978369$, $u/H = -2.60176$, against 1 and $\frac{4}{3}$ on the history); over 38 cases with three further theories the outcomes are a_4 turning 24 times, regular 6 times, runaway 4 times and singular 4 times. These are statements of that approximation, not of the field equations.

Consequence: the Kohn-Sham gas of this record is a test field on the prescribed background $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$ without back-reaction; energies, pressures and equations of state along the history are not consequences of the coupled field equations.

report	records
`Revision/field_equations_a4/reports/ks-source-conditions.json`	`ks_profiles_depend_on_x0`, `ks_hidden_pressure_condition`, `ks_offdiagonal_condition`, `ks_history_is_a_prescribed_background`, `ks_zero_source_states_listed`

report	records
Revision/field_equations_a4/ks_source/reports/ks-source-a4.json	A1_hidden_equals_extra_times_and_no_offdiagonal, A2_constraint_first_order_second_order_system, A3_constraint_propagation, A4_x0_conservation_in_y, A5_free_fold_band_gas_conserves_energy, A6_history_source_reproduced, B0_source_condition_report_verified, B1_integrals_reproduced, B2_x0_conservation_on_every_profile, B3_hidden_direction_mismatch, C1_averaged_hidden_pressure_condition, C2_energy_relation_along_the_dense_history, C3_record_history_not_sourced_by_the_gas, C4_source_signs, D0_free_gas_average_exact, D1_two_step_sizes_agree, D2_author_weights_from_the_record_history, D3_constraint_kept_along_the_solutions, D4_outcomes
Kohn-Sham theory (the record and the label of the history)	adiabatic_offdiagonal_identity, ks_theory_json_written, ks_theory_json_basis, ks_theory_json_slope, ks_theory_json_history_label

12 The Kohn-Sham-level pairing T3

The block map of the chirality Γ, $(\chi, j) \to (\sigma_2\chi, -j)$, with the two fold conditions exchanged and the tip angle $\theta \to \pi - \theta$, maps every self-consistent instantaneous Kohn-Sham state with (m, λ, θ) onto one with $(-m, +\lambda, \pi-\theta)$, with equal levels, occupations, Kohn-Sham energies and energy-momentum profiles and $S \to -S$ (both fold conditions filled, ASSUMED; mean-field level); a_4 and b_4 enter h_j only through the constants f and κ of the slice, so it holds slice by slice. The theorem, its hypotheses and its proof are in `Revision/docs/PAIR_CREATION_PROOFS.md` section 8, verified by `Revision/pairing/kohn_sham/reports/wolfram-t3.json` (9 of 9 PASS) and `Revision/pairing/kohn_sham/reports/python-t3.json` (12 of 12 PASS); its completion record (`Revision/pairing/kohn_sham/t3-completion.json`) by `Revision/pairing/kohn_sham/reports/wolfram-t3-completion.json` (3 of 3 PASS) and `Revision/pairing/kohn_sham/reports/python-t3-completion.json` (7 of 7 PASS). The Rust solver contains a numerical self-test, labelled in its report as NOT a proof of T3: $(m, \lambda, \theta = 0)$ and $(-m, \lambda, \theta = \pi)$, solved independently, agree with opposite S (worst 3.62e-13, tolerance 1e-9; record `t3_block_map_solver_selftest`). T3 does not establish a creation process, a rate or an amplitude.

13 Verification records

13.1 Reports and their counts

Every check has a name, a verdict and a detail. Counts at the time of writing (the publication test re-reads them):

report	checks	PASS	FAIL
Revision/kohn_sham/reports/ks-theory-wolfram.json	51	51	0
Revision/kohn_sham/reports/ks-theory-python.json	61	61	0
Revision/kohn_sham/reports/ks-rust-solver.json	45	45	0
Revision/kohn_sham/reports/ks-rust-determinism.json	14	14	0
Revision/kohn_sham/reports/ks-rust-mermin-roots.json	5	5	0
Revision/kohn_sham/reports/ks-reference.json	39	39	0
Revision/kohn_sham/reports/ks-crosscheck.json	31	31	0
Revision/field_equations_a4/reports/ks-source-conditions.json	5	5	0
Revision/field_equations_a4/ks_source/reports/ks-source-a4.json	19	19	0
Revision/field_equations_a4/reports/wolfram-a4-report.json	57	57	0
Revision/field_equations_a4/reports/python-a4-report.json	69	69	0
Revision/pairing/kohn_sham/reports/wolfram-t3.json	9	9	0

report	checks	PASS	FAIL
Revision/pairing/kohn_sham/reports/python-t3.json	12	12	0
Revision/kohn_sham/tip_convergence/tip-convergence.json	7	7	0

13.2 The independent reference and the full cross-check

The reference shares no code with the Rust solver: different discretisation (staggered finite differences with Richardson extrapolation instead of RK4 shooting), its own SCF and continuation fallback, its own parameter derivation (it re-derives the particle numbers and couplings from the stated rules) and its own uncertainty, validated on a fourth grid (record `richardson_uncertainty_validated`). The checker `Revision/kohn_sham/checker/crosscheck_ks.py` compares the two on the FULL canonical matrix: 75 ground states, 135 thermal states, 60 exact-Fock-variant states, 60 rescaling partners, 75 particle-hole lists and the 5 slices of the crossing demonstration, 217571 comparisons in all, under a tolerance rule fixed before the comparison,

$$|x_{\mathrm{Rust}} - x_{\mathrm{ref}}| \le 3\,(U_{\mathrm{ref}} + U_{\mathrm{Rust}}) + 10^{-12}\,\mathrm{scale}, \qquad U_{\mathrm{Rust}} = \tfrac{16}{15}\,|\mathrm{canonical} - \mathrm{refined}|,$$

with U_{ref} the reference's three-grid Richardson uncertainty and U_{Rust} measured for every state of the matrix by the refinement measurement, whose script and record are `Revision/kohn_sham/checker/measure_rust_refinement.py` and `Revision/kohn_sham/checker/rust-refinement.json`. All 31 checks of the cross-check pass:

check	check	check
inputs_all_pass	rust_refinement_applies_to_matrix	rust_refinement_applies_to_matrix_exx
problem_definition_identical	parameters_particle_numbers	parameters_couplings
ground_occupations_and_groups	ground_energies	ground_homo_lumo_gap
ground_eigenvalues	excited_delta_scf	excited_particle_hole_lists
emt_integrals	emt_brane_tip_values	ground_profiles
exchange_delta_E_x	adiabatic_dE_da4	adiabatic_Q_max
exx_variant_scf	rescaling_partners	crossing_demonstration
thermo_state_functions	thermo_derivatives	thermo_sea_hole_diagnostic
thermo_levels	thermo_mu_high_precision	thermo_mu_rounding_diagnostic
thermo_mu_rust_stated_bound	reference_outputs_lf_only	reference_manifest
reference_repeat_byte_identical		

13.3 Determinism and the refinement measurement

The Rust determinism report (14 of 14 PASS): a repeat of the canonical matrix is byte-identical in all 244 result files and in its check report; a run with refined tolerances (900 to 1800 RK4 steps, root tolerance 1e-14, SCF tolerance 1e-14 with at most 1000 iterations) agrees with the canonical run to 1.728e-11 in E_{KS} (relative), 5.688e-10 in 50420 levels, 1.950e-10 in the profiles, 4.513e-08 in $Q_{\max}$ and dE/da_4, 1.874e-10 in the thermodynamics and 3.853e-09 in C_V. The refined run has the SCF tolerance 1e-14, three orders below the canonical one, so that the canonical-refined differences contain the canonical SCF stopping error (with an SCF tolerance only ten times smaller both runs could stop at the same iteration and carry the same stopping error; `claude_failures.md` H15).

item	records of `Revision/kohn_sham/reports/ks-rust-determinism.json`
the repeat	`repeat_byte_identical`, `repeat_report_byte_identical`, `outputs_lf_only`
the refined run	`refined_same_inputs`, `refined_free_spectra_convergence_order`, `refined_ground_energies`, `refined_ground_homo_lumo_gap`, `refined_eigenvalues`, `refined_delta_scf`, `refined_profiles`, `refined_adiabatic_derivatives`, `refined_thermodynamics`, `refined_mermin_root_path`, `refined_heat_capacity`

13.4 Re-verification for this document

For this document (2026-10-17, the corrected metric) the committed Rust binary was run on the canonical matrix again into a scratch work folder, in the three parts and the assembly of the gate (`revision_ks_solver all --cache <scratch> -part i/3 -inner-threads 2`, then `-out <scratch>/results -report <scratch>/report.json`; 76.7 s, 70.4 s, 85.7 s and 5.7 s): all 244 result files and the check report were byte-identical to the committed `Revision/kohn_sham/results` and `Revision/kohn_sham/reports/ks-rust-solver.json` (45 of 45 checks PASS). The whole Kohn-Sham record (the repeat, the refined run, the reference, its repeat, the cross-check, the tip-cutoff study, the dense history and the source analysis) was regenerated the same day in 57 steps, each under the time limit of 423 s for one calculation, every step with exit code 0 (`claude_failures.md` H15 and H16). The other reports cited are the files of the working tree; their counts are re-read by the publication test.

14 Reproduction

From the repository root, in this order (the Kohn-Sham theory record reads the gammas of `Revision/algebra`; the Rust solver reads `ks-theory.json`; the Mermin-root tool runs after the canonical matrix; the reference runs in six parts per stage with a job cache, each part within the limit of 423 s for one run, and a final run writes its outputs; the cross-checker runs after the reference and the

refinement measurement, with its own repeat of the reference, also in parts; the source-conditions checker reads the Kohn-Sham results and the Kohn-Sham theory checker reads its record; the Kohn-Sham source analysis reads the dense free-gas history of the dark sector, `Revision/dark_sector/dirac16complex/compute/run_ks_history.py`; `<scratch>` is any empty directory outside the repository; the backslash is the line continuation of a POSIX shell; the gate scripts `Revision/verify_revision.sh` and `Revision/verify_revision.ps1` run the same steps in parts):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/kohn_sham/theory/verify_ks_theory.wls
cargo build --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
cargo test --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
Revision/kohn_sham/solver/target/release/revision_ks_solver all \
    --out Revision/kohn_sham/results \
    --report Revision/kohn_sham/reports/ks-rust-solver.json
python Revision/kohn_sham/solver/tools/mermin_roots_mp.py --work <scratch>/mermin-roots
Revision/kohn_sham/solver/target/release/revision_ks_solver all \
    --out <scratch>/repeat --report <scratch>/repeat-report.json
Revision/kohn_sham/solver/target/release/revision_ks_solver all --refined \
    --out <scratch>/refined --report <scratch>/refined-report.json
python Revision/kohn_sham/solver/tools/compare_runs.py \
    --canonical Revision/kohn_sham/results \
    --canonical-report Revision/kohn_sham/reports/ks-rust-solver.json \
    --repeat <scratch>/repeat \
    --repeat-report <scratch>/repeat-report.json --refined <scratch>/refined \
    --refined-report <scratch>/refined-report.json \
    --report Revision/kohn_sham/reports/ks-rust-determinism.json
python Revision/kohn_sham/reference/run_reference.py --cache <scratch>/cache \
    --stage 1 --part 1/6 --jobs 20   # and so on up to --part 6/6, then --stage 2
python Revision/kohn_sham/reference/run_reference.py --cache <scratch>/cache --jobs 20
python Revision/kohn_sham/checker/measure_rust_refinement.py \
    --work <scratch>/rust-single --jobs 22
python Revision/kohn_sham/reference/run_reference.py --cache <scratch>/repeat-cache \
    --stage 1 --part 1/6 --jobs 20   # and so on, both stages
python Revision/kohn_sham/checker/crosscheck_ks.py --work <scratch>/cc --jobs 22 \
    --repeat-cache <scratch>/repeat-cache
python Revision/kohn_sham/tip_convergence/tip_convergence.py \
    --work <scratch>/tip --jobs 22 --stage scan --part 1/6   # parts, recal, final
python Revision/field_equations_a4/python/check_ks_source_conditions.py
python Revision/kohn_sham/theory/check_ks_theory.py
python Revision/dark_sector/dirac16complex/compute/run_ks_history.py \
    --solver Revision/kohn_sham/solver/target/release/revision_ks_solver --part 1/3
python Revision/field_equations_a4/ks_source/ks_source_a4.py
python -m unittest Revision/field_equations_a4/ks_source/test_ks_source_a4.py -v
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3.wls
python Revision/pairing/kohn_sham/python/check_t3.py
python scripts/build_provenance_pdf.py Revision/docs/KOHN_SHAM_DEFLATING_FIELD.md \
    --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_kohn_sham_deflating_field_publication.py -v
```

Expected output: every verifier ends with all of its checks passing and writes the report of section

13.1 with the counts given there (the Wolfram Kohn-Sham verifier prints 51/51 checks passed; the Rust solver prints SUCCESS after 45 PASS lines; the cross-checker writes 31 of 31 PASS; the tip-cutoff study prints 7 PASS lines and SUCCESS); every run is deterministic, and a second run gives byte-identical outputs. Run times recorded in the READMEs of the folders (longer when the machine is shared): the Wolfram Kohn-Sham verifier about 40 to 85 s and the sympy checker about 75 to 165 s (`Revision/kohn_sham/theory/WOLFRAMSCRIPT_PROVENANCE.md`); on 2026-10-16/17 with 22 threads the Rust canonical matrix in two parts of 134.6 s and 130.5 s with an assembly of 5.4 s, the repeat in three parts of 76.7 s, 70.4 s and 85.7 s, the refined-tolerance run in four parts of 176.5 s, 186.1 s, 183.0 s and 163.8 s (`Revision/kohn_sham/solver/README.md`); the reference in two stages of six parts, stage 1 193.0, 175.4, 214.0, 179.8, 100.7 and 191.2 s, stage 2 136.8, 143.7, 133.7, 133.7, 134.2 and 139.8 s (`Revision/kohn_sham/reference/README.md`); the refinement measurement 108.8 s and the cross-checker itself 87.6 s (`Revision/kohn_sham/checker/README.md`); the tip-cutoff study 966 s in ten parts (`Revision/kohn_sham/tip_convergence/README.md`); the Kohn-Sham source analysis about 4.4 s and its unit tests about 20 s (`Revision/field_equations_a4/ks_source/README.md`); the a_4 verifiers 20 s (Wolfram) and 8 s (sympy) (`Revision/field_equations_a4/README.md`); the T3 verifiers 5.8 s and 1.3 s (`Revision/pairing/kohn_sham/wolfram/WOLFRAMSCRIPT_PROVENANCE.md`).

The PDF build runs the Markdown-to-LaTeX builder twice (with different hash seeds), runs pdflatex three times into each of two fresh directories, requires warning-free logs and byte-identical PDFs, and checks the PDF against its entry `kohn-sham-deflating-field` in `Revision/pdf-specifications.json` (the Revision registry; the registry of the earlier stages is not touched).

15 What is proved, computed, assumed and not established

15.1 Proved (exact, two independent verifiers)

1. The block reduction (section 3): for every history $a_4(x_4)$, $b_4(x_4)$ the good-sector equation reduces exactly to eight 2×2 blocks h_j in the hidden coordinate; the spin-connection terms $\frac{3}{2}(H/f)\gamma^0$ and $(b_4'/b_4 - \frac{1}{2}a_4')\gamma^4$ are removed by the half-density factor $e^{-(3/2)Hy}e^{a_4/2}b_4^{-1}$; the hidden direction enters through $1/f$ and the 3-momenta through the constant $\kappa = e^{-a_4}$.

2. The exchange functional (section 4): the exact local exchange of the uniform 8-fold good-sector gas at every temperature, $M_{\text{eff}} = m + \frac{15}{16}\lambda S$, $v_v = -\lambda n/16$; the exact Fock exchange of the closed-shell slab differs by $+\lambda Q^2/32$; a state depends on the coupling and the cell volume only through their ratio.

3. The boundary-value structure (section 5): self-adjointness with current-killing ends, the fold conditions, the exact $k = 0$ spectra, the fold band $\varepsilon = \kappa k$ and its slope.

4. The rescaling identity (section 10), the energy-momentum tensor of the Kohn-Sham states with its trace and conservation identities and the energy change along the history (section 9), and the adiabaticity identity of the measure (section 7).

5. T3 (section 12; proved in `Revision/docs/PAIR_CREATION_PROOFS.md`), under its stated hypotheses.

15.2 Computed (numerical, two independent solvers, cross-checked)

1. The instantaneous ground states, first excited states (gap, Δ-SCF, particle-hole lists), Mermin states, energy-momentum profiles, rescaling partners and exact-Fock variants of the canonical matrix (sections 6 to 10), with the Rust solver and the reference agreeing on the full matrix within the stated uncertainty (31 of 31 checks, 217571 comparisons).

2. The adiabaticity measure ($Q_{\max} \leq 0.1982205$) and the absence of Fermi-level crossings in the canonical matrix; the crossing demonstration at $N = 2360$ (section 7).

3. That the recorded states are not admissible sources of the a_4 and b_4 equations (section 11).

4. The dependence on the tip cutoff L from 3 to 6 of 29 recorded states, the 27 ground states with $N \in \{8, 592, 2352\}$, $\lambda \in \{0, \pm\lambda_1\}$, $a_{4,0} \in \{0, 1, 2\}$ and two Mermin states of $N = 592$ ($\lambda = 0$, $a_{4,0} = 2$, $T = 0.05$; $\lambda = \lambda_1$, $a_{4,0} = 1$, $T = 0.02$) (section 5; `Revision/kohn_sham/tip_convergence/`, 7 of 7 checks PASS; the Rust solver at every L, the reference at $L = 3$ and 4).

15.3 Assumed or chosen

1. Both fold conditions filled at the fold $z = \pi/2$, where the two halves of a cell meet (ASSUMED); the metric is degenerate there and no junction condition is derived.

2. The regular tip at the cutoff $L = 3$ with $\theta = 0$ (chosen). Its effect is measured (section 5): the free fold-band gases do not depend on L, except their values at the fold (the normalisation of the fold-band orbital) and except $N = 2352$ at $a_{4,0} = 0$; the interaction energies do: the recorded $N = 8$ interaction energy is 0.977, 0.674 and 0.215 of its large-L value at $a_{4,0} = 0, 1, 2$. The couplings (calibrated at $L = 3$) and $N = 2352$ (bulk edge at $L = 3$) are themselves $L = 3$ choices.

3. The filling convention (particles on the positive branch and on the $k = 0$ zero modes; the sea not populated thermally), whose justification is OPEN; the sea-hole diagnostic shows where it fails (27 of 135 thermal states).

4. The history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$, $A = 1$, PRESCRIBED, not solved for.

5. The parameters of section 5 (N, $\hat{\lambda}$, L, Δk, v_t, T).

15.4 Not established

1. The time-dependent (non-adiabatic) Kohn-Sham problem (TDDFT) is OPEN: only instantaneous states are computed; the measure $Q_{\max}$ is a consistency indicator, not a solution of the evolution.

2. Correlation: the functional is Hartree plus exchange only; no correlation functional is derived or used.

3. Back-reaction: the Kohn-Sham states are a test field on a prescribed background; no recorded state is an admissible source of the field equations for a_4 and b_4, and no history $a_4(x_4)$, $b_4(x_4)$ is derived from the Kohn-Sham source.

4. Equations of state and the dark-sector hypothesis are not part of this document (`Revision/dark_sector/dirac16complex`).

5. No creation process, rate or amplitude for universes follows from the Kohn-Sham model or from T3.

6. A thermal treatment of the Dirac sea (pairs) where the particle-only ensemble fails; states beyond the computed range $a_{4,0} \in [0, 2]$; the continuum limit $\Delta k \to 0$; Kohn-Sham states with extra-time momenta. The limit $L \to \infty$ is established (numerically, with uncertainties) only for the states and quantities marked converged in `Revision/kohn_sham/tip_convergence/README.md`; it is NOT established for $N = 2352$ at $a_{4,0} = 0$ (the $k = 0$ bulk levels approach m algebraically and the occupied set changes; the iteration of its interacting partners fails), nor for the Mermin state N592_lam0_a20_T50 (successive changes of about 10^{-8} relative that do not meet the geometric-tail rule). It is not computed at $L \neq 3$ for λ_2, the slices 0.5 and 1.5, $T = 0.01$ and the other thermal states, the excited states and the adiabaticity measure, nor at $L > 6$. The record's calibration rule drives λ_1 to zero as L grows.